

YSU Quantitative Finance

Spring 2026, YSU

Project No. 02

Due time/date: 11:30, 17 June, 2026

Project 2a: Supplementary

Open a paper account in Interactive Brokers or Alpaca, and create a portfolio.

Project 2b: Supplementary

Dashboard for Bayesian updates

Create an interactive dashboard for the Bayesian updates for the following cases:

Bernoulli Data: $X \sim \text{Bernoulli}(p)$

User chooses the prior distribution of p . Prior is given by $\text{Beta}(a, b)$, and user chooses a and b . The default values of a and b are $a = 1, b = 1$, resulting in uninformative (Uniform) prior. The dashboard plots the Prior's PDF. Then user enters the data (say, 1100101100), and the dashboard plots the Posterior's PDF. Also, the dashboard needs to show the MAP and Bayes Estimates for p using that data.

Normal Data: $X \sim \mathcal{N}(\mu, \sigma^2)$

This is for the distribution of the Normal Mean μ . First, user chooses the value of σ , with the default of $\sigma = 1$. Then user chooses the prior distribution for $\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$, by choosing μ_0 and σ_0 . The default values are $\mu_0 = 0$ and $\sigma_0 = 1$, resulting in Standard Normal prior. The dashboard plots the Prior's PDF. Then user enters the data (say, 0.123, 1.3812, -0.13), and the dashboard generates plots the Posterior's PDF. Also, the dashboard needs to show the MAP and Bayes Estimates for p using that data.

Supplementary: You can add a feature to have the following: user assumes that he knows the true value for μ , she enters μ . Then she gives μ_0, σ_0 , and enters the number of points to generate from the true distribution. Then the dashboard generates random points from the true distribution, and draws the posterior using that points. the dashboard shows the true distribution PDF, the Prior distribution PDF and the Posterior. This is to interactively show that when the number of points from the true distribution increases, the posterior tends to the true distribution, as the number of points increases, and the prior distribution/information is not so important as $n \rightarrow +\infty$.

Project 2c:

Project Overview

Each group will select:

- several stocks, or
- a stock index and several constituent stocks, or
- stocks from a particular industry sector.

Historical daily data should cover at least 2-5 years.

Part I – Linear Regression Models for Financial Data

In this part you can work on the following problems (you are free too choose any subset of this and also other similar questions):

- Can one stock's returns explain another stock's returns?
- Can market index returns explain individual stock returns?
- Can past returns help predict future returns?
- Can trading volume explain returns?

You can split the problem(s) into several stages:

Stage 1. Simple Linear Regression

Here you need to construct a simple regression model for the above problems, for example:

Examples:

$$\text{Stock Return}_t = \beta_0 + \beta_1 \cdot (\text{Market Index Return})_t + \varepsilon_t$$

or

$$\text{Stock Return}_t = \beta_0 + \beta_1 \cdot (\text{Market Index Return})_{t-1} + \varepsilon_t$$

or

$$\text{Stock Return}_t = \beta_0 + \beta_1 \cdot (\text{Stock Return})_{t-1} + \varepsilon_t$$

Stage 2. Multiple Linear Regression

In second stage, you are asked to construct a multiple regression model involving several predictors for above selected problems.

Say, you can try to predict, by Multiple Linear Regression, the return of a particular stock, using (some of) the following (or other) predictors:

- market index return (at time t or $t - 1$ or other),
- lagged return of the same stock ($r_{t-1}, r_{t-2}, \dots$),
- moving average,
- volatility,
- trading volume.

Stage 3. Model Evaluation and selection

For each of the obtained Linear Regression Models, please report the followings:

- coefficient estimates and the obtained model
- significance tests for coefficients
- R^2 and the adjusted R^2 Coefficients
- (supplementary) residual diagnostics.

Please also do some kind of model selection: among different multiple linear regression models, choose the one which best explains and/or predicts the explained variable. Discuss which predictors matter.

Part II – Bayesian Analysis of Stock Returns

Here you need to use the Bayesian reasoning to update beliefs about unknown quantities based on observed data. For example, you can consider the problem of estimating the probability that the daily return of some stock or index will be positive. Assuming that the returns are independent, this can be modeled by the Bernoulli distribution with the unknown probability of success (the return being positive) p . Or, assuming that the returns are Normally Distributed (which, as we saw earlier, is not a reasonable assumption, but, still, easy to handle with), you can estimate the mean μ .

Please do some visualizations, using the above Dashboard or just some simple pictures.

Part III – Markov Chain Models of Market Behavior

Here you need to consider a problem of a Markov Chain type. Say, you can consider the Market Regimes problem: we consider the following states (regimes) for the market: Bull, Neutral and Bear. First, you need to give the definition of these regimes (say, you can consider the S&P500 index or the NASDAQ index, and if it is positive for 5 consecutive trading days, then the market is in Bull state). Now, you need to estimate the transition probability matrix, and, say, predict the next most probable state, estimate the proportion of time market spends in each state or find the stationary distribution of states (if exists). Or, you can create your own portfolio, and do the same thing (define portfolio states as Bull, Neutral and Bear) for your portfolio movements.

Also, you can use the estimated Markov chain to simulate future market states, and compare the simulated and observed behaviors.